

From Sensor Responses to an Output-Only Information Bound

Friday 25 September 2026: SSI-bridge findings, and the plan for Saturday 26 and Sunday 27 September 2026

Technical writeup, continued

This note records Friday’s work on the SSI bridge: the step from the EMM forward model to features that can be measured from floe accelerations alone. It follows the EMM inversion report, whose open item was an SSI-compatible feature vector. The day began by testing the SSI chain on broadband synthetic data (Section 3), then measured what the sensor responses actually look like (Section 4) and how much they depend on each parameter (Section 5), fixed γ on that evidence (Section 6), and finally quantified how much parameter information survives the loss of the input (Sections 7 and 8). The weekend plan is in Section 11. Colour convention as before: findings in green, open or scope-limited points in red, definitions and theory in blue.

1 Setting and notation

Forward model. The EMM (Montiel) solves the time-harmonic problem for a circular elastic plate of non-dimensional radius R , flexural rigidity $\beta = D/(\rho_0 g H^4)$ and draught $\gamma = d/H$ in water of depth $H = 1.88$ m, forced by a plane incident wave travelling along the x -axis. Lengths are scaled by H and the frequency enters through

$$\alpha = \frac{H\omega^2}{g}, \quad \omega \text{ in rad/s.} \quad (1)$$

The time convention is $\zeta(r, \theta, t) = \text{Re}\{\eta(r, \theta) e^{+i\omega t}\}$ (Montiel, Eqs. 2.9–2.10), the same as `evaluateSpectralDeflection.m`. The reference point is $p_0 = (\beta_0, \gamma_0, R_0) = (4.6985 \times 10^{-5}, 1.4548 \times 10^{-3}, 0.383)$. The truncation $[M, P, N] = [50, 10, 10]$ is validated for $\alpha \in [1.7, 13.85]$, i.e. $\omega \approx 2.9$ – 8.5 rad/s; this is the *trusted band*.

Sensor frequency response functions (FRFs). For sensor j at (r_j, θ_j) the displacement FRF is the plate deflection per unit incident-wave amplitude, $H_{\eta,j}(\omega) = \eta(r_j, \theta_j; \omega)$ (dimensionless: metres per metre). An IMU measures acceleration, so the observable is

$$H_j(\omega) \equiv H_{a,j}(\omega) = -\omega^2 H_{\eta,j}(\omega) \quad [\text{ms}^{-2} \text{ per metre of incident amplitude}]. \quad (2)$$

The factor $-\omega^2$ leaves the poles and the normalised spatial vector $H/\|H\|$ unchanged (up to sign) but changes where broad maxima sit. The four-sensor vector is $\mathbf{H}(\omega) = [H_1, H_2, H_3, H_4]^T$.

Sensor	Candidate index	r/R	θ	$x = r \cos \theta$ (non-dim.)
s2	2	0.3	0	+0.1149
s6	6	0.3	π	−0.1149
s14	14	0.5	π	−0.1915
s26	26	0.9	0	+0.3447

Table 1: Sensor layout: the Level 2C long-horizon configuration [2 6 14 26] of `level2C_finalCheck.m`, used as a named baseline (not an optimum for this problem). All four lie on the incidence axis, so only responses symmetric about that axis are observed, which is everything a single incident direction excites.

2 Local information framework

Every comparison in this note uses one local (Gauss–Newton, Fisher) framework, so the numbers in different sections are directly comparable.

Log-parameters and derivatives. The parameters live on very different scales, so all derivatives

are taken with respect to $\ln p_i$ by central differences,

$$\frac{\partial \mathbf{H}}{\partial \ln p_i} \approx \frac{\mathbf{H}(p_i e^\epsilon) - \mathbf{H}(p_i e^{-\epsilon})}{2\epsilon}, \quad \epsilon \in \{10^{-2}, 10^{-3}\}, \quad (3)$$

with $\epsilon = 10^{-3}$ primary and 10^{-2} as a step check. After the decision of Section 6 the inverse parameter vector is $\theta = [\ln \beta, \ln R]^T$.

Noise surrogate. The reference observable is the complex FRF written as real components $[\text{Re } \mathbf{H}(\omega_k); \text{Im } \mathbf{H}(\omega_k)]$ on a uniform grid of spacing $\Delta\omega$. Each real component is modelled as independent with variance

$$\sigma_k^2 = \sigma_n^2 \frac{\|\mathbf{H}(\omega_k)\|^2}{\Delta\omega}, \quad (4)$$

where σ_n is a *relative FRF noise density* (per $\sqrt{\text{rad/s}}$). The factor $1/\Delta\omega$ makes the information an integral over frequency, independent of how densely the band is sampled. This is a structural surrogate, not the covariance of any real estimator; that comes in the weekend work (G2).

Information matrix and parameter uncertainty. For an observable with parameter Jacobian J and noise covariance Σ , the local information matrix is

$$F = J^T \Sigma^{-1} J \approx \int J(\omega)^T \Sigma(\omega)^{-1} J(\omega) d\omega. \quad (5)$$

Two standard deviations are reported, both per unit σ_n :

$$\sigma_i^{\text{joint}} = \sqrt{[F^{-1}]_{ii}} \quad (\text{all parameters estimated together}), \quad \sigma_i^{\text{fixed}} = \frac{1}{\sqrt{F_{ii}}} \quad (\text{the others known}). \quad (6)$$

Their ratio measures coupling. The singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3$ of the whitened stacked Jacobian give the condition number $\kappa = \sigma_1/\sigma_3$ and the weakest parameter direction $v_{\min}$ (last right singular vector). A one-standard-deviation target of 10% relative accuracy (0.1 in $\ln p$) therefore requires $\sigma_n \lesssim 0.1/\sigma^{\text{joint}}$.

Information retained by a feature set. Let $F_f = J_f^T \Sigma_f^{-1} J_f$ be the information in a candidate feature set and F_r that of the reference. The generalised eigenproblem

$$F_f v = \lambda F_r v \quad (7)$$

is solved symmetrically through $F_r = LL^T$ (λ are the eigenvalues of $L^{-1}F_f L^{-T}$). Each λ_i is the fraction of the reference information kept along direction v_i (1: nothing lost; 0: everything lost). Information is matrix-valued, so no single percentage is quoted. If the features are a locally linear function of the reference, $f \approx Ay$, and their noise is propagated consistently, $\Sigma_f = A\Sigma_r A^T$, then necessarily $0 \leq \lambda \leq 1$; $\lambda > 1$ flags inconsistent noise models. Directions are reported in units of the reference standard deviations (v_i divided elementwise by σ_r^{joint} , renormalised), because the raw v_i is always dominated by whichever parameter is least well known.

3 Morning: broadband SSI on a known oscillator (Level 4A)

Before touching the EMM, the synthesis-plus-SSI chain was tested on analytical damped modal systems (one mode, $\omega_n = 6$, $\zeta = 0.05$; two close modes at 5.5 and 6.3 rad/s with $\zeta = 0.04, 0.06$), observed as acceleration, 30 realisations per case.

Frequency-domain stochastic synthesis (`stochasticSensorSynthesis.m`). With one shared

complex Gaussian input coefficient ξ_k per frequency bin,

$$A_k = w_k \sqrt{S_u(\omega_k) \Delta\omega} \mathbf{H}(\omega_k) \xi_k, \quad y(t) = \text{Re} \sum_k A_k e^{i\omega_k t}, \quad (8)$$

assembled with explicit Hermitian symmetry and an inverse FFT. Sharing ξ_k across sensors keeps the spatial coherence (the cross-spectrum is rank one, $S_u w^2 \mathbf{H} \mathbf{H}^H$). The record is exactly one period $2\pi/\Delta\omega$; $N_{\text{BW}} = 2\zeta\omega_n/\Delta\omega$ counts bins across the half-power band.

Exact band limitation breaks SSI; an innovation floor restores it. With $\Delta t = 0.1$ s the sampled band extends to 31.4 rad/s, but the synthesis excited only 3–9 rad/s. A stationary process whose spectrum vanishes on a set of positive measure has $\int \ln S = -\infty$, so by the Szegő–Kolmogorov formula its one-step prediction error is zero: it is deterministic, and SSI fits it with near-undamped poles. Without noise the modes were not detected at any $N \geq 256$; with a white measurement floor of 10^{-3} to 10^{-1} of the signal standard deviation they were detected in essentially every realisation. The full time-domain control recovered both close modes from $N = 2048$.

Quantity	Single mode	Close mode 1	Close mode 2
Gate A: $\hat{\zeta}/\zeta$ (synthesis) minus (band-passed time domain), pooled two longest N	+0.069	+0.174	−0.004
Band [3, 9] vs full band: $\hat{\zeta}/\zeta$ at $N = 4096$	0.93 → 0.94	1.02 → 0.99	0.80 → 0.92
Band [3, 9] vs full band: persistent spurious branches	3.7 → 0.1	3.5 → 0.3	3.5 → 0.3

Table 2: Level 4A summary (MATLAB run `level4A_20260925_110856`). The Gate A criterion 0.15 was fixed in advance and is reported unchanged: narrowly missed for close mode 1; 0.20 would pass all three. The band-width rows are qualitative (band, realisation and absolute noise co-varied).

What Level 4A did not test. The excitation was white. Coloured excitation (a real sea state) was not tested, and Section 8 shows why it matters.

4 What the sensor FRFs look like (file 5)

`computeSensorFRF.m` wraps one EMM solve per frequency (`precomputeDeflectionData`) and evaluates all sensors from it (`evaluateDeflection`); its tester (10 groups) confirms exact agreement with direct calls, $H_a = -\omega^2 H_\eta$ exactly, mirror symmetry $\eta(r, \theta) = \eta(r, -\theta)$ to 5.5×10^{-16} , and about 1.03 s per solve in MATLAB. `runSensorFRFReconnaissance.m` then swept 221 frequencies over 3–8.5 rad/s at p_0 .

Affine (rigid-like) split. Because all four sensors lie on the x -axis, each frequency is fitted by least squares as

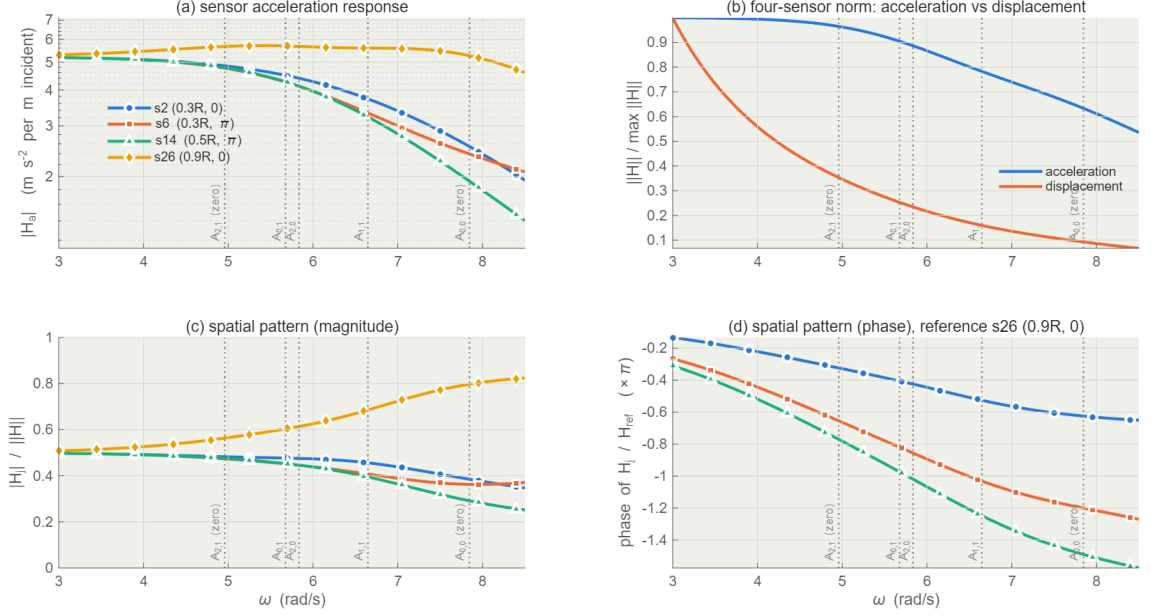
$$H_j(\omega) = c_0(\omega) + c_1(\omega) x_j + E_j(\omega), \quad (9)$$

where c_0 is *heave-like*, c_1 *pitch-like* (least-squares coefficients, not motions measured at the floe centre) and E is the non-affine remainder (flexure plus the floe following the curvature of the incident wave). The affine projector $P_A = X(X^T X)^{-1} X^T$, $X = [\mathbf{1}, \mathbf{x}]$, and $I - P_A$ are orthogonal, so $\|P_A \mathbf{H}\|^2 + \|(I - P_A) \mathbf{H}\|^2 = \|\mathbf{H}\|^2$.

No isolated sensor-level resonances. $\|\mathbf{H}\|$ has no peak above 0.3 dB prominence; three sensors are monotone and s26 has a single 0.6 dB bump at 5.4 rad/s that never reaches a half-power width. Relative phases follow the incident-wave propagation $k(x_j - x_{\text{ref}})$ at low frequency and fall short of it at high frequency (s2: -0.133π vs -0.126π at 3 rad/s, -0.651π vs -1.01π at 8.5 rad/s). The structure sits in the affine split (Table 3), and it is broad.

Curve	Feature	Frequency (rad/s)	Size
$ c_1 $ (pitch-like)	peak	5.475	6.4 dB, half-power width 3.43 rad/s
$ c_0 $ (heave-like)	dip	7.275	3.9 dB
$\ E\ $ (non-affine)	peak	7.125	1.8 dB, not isolated
$\ E\ /\ \mathbf{H}\ $	share	3.0 / 5.75 / 8.5	0.043 / 0.382 / 0.670

Table 3: Reconnaissance structure at p_0 (sensorFRFrecon_20260925_122410). Truncation check [50 10 10] vs [60 12 12]: relative change 1.3×10^{-5} , 7.6×10^{-5} , 2.7×10^{-4} , 7.3×10^{-4} at 4.0, 5.5, 7.0, 8.3 rad/s, so the smoothness is not a truncation artefact.



Shaded: proposed pole windows. Dotted lines: old EMM features ("zero" = near-zero feature); overlay only, not expected to coincide.

Figure 1: Sensor acceleration FRFs at p_0 : (a) per sensor, (b) four-sensor norm, acceleration vs displacement, (c) spatial magnitude pattern, (d) phase relative to s26. Dotted lines mark the old dry-mode features (overlay only).

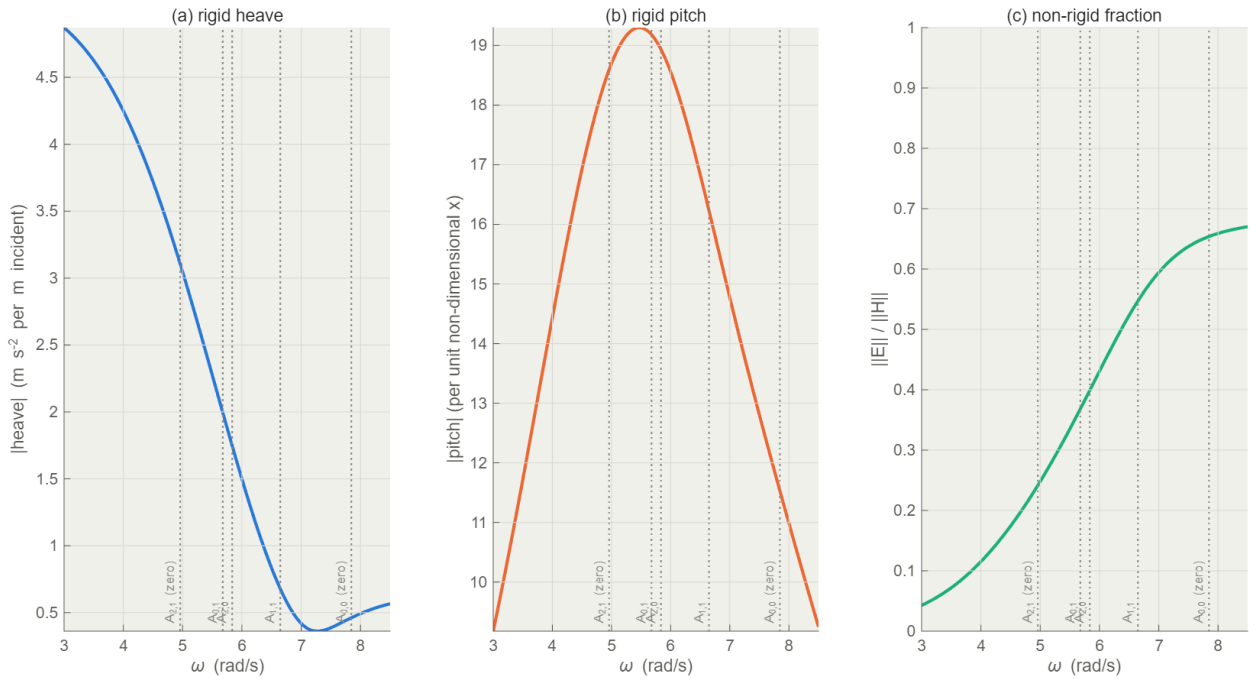


Figure 2: Affine split: (a) heave-like $|c_0|$, (b) pitch-like $|c_1|$, (c) non-affine share $\|E\|/\|\mathbf{H}\|$.

Consequence. The original plan (fit a local pole to each resonance window) has nothing to fit. Why the response is smooth is not established here; heavy hydrodynamic damping, forcing and numerator effects, and overlapping contributions are all candidates.

5 How much each parameter changes the sensor FRFs (file 5b)

`runSensorFRFSensitivity.m` computed $\partial \mathbf{H} / \partial \ln p$ for $p = (\beta, \gamma, R)$ at 45 frequencies over the trusted band, for three representations of the same response: raw sensors, the affine part $P_A \mathbf{H}$ and the non-affine part $(I - P_A) \mathbf{H}$. Sensors stay at fixed coordinates when R is perturbed. The step check was clean: the two steps agree to 1.58×10^{-5} (β), 1.85×10^{-5} (γ) and 4.78×10^{-4} (R).

Frequency-resolved sensitivity. For representation P , $S_i(\omega) = \|P \partial \mathbf{H} / \partial \ln p_i\| / \|\mathbf{H}(\omega)\|$: the relative change of the response per unit relative change of p_i .

ω (rad/s)	3.00	3.88	4.88	5.75	6.62	7.62	8.50	rms
S_β	8.1×10^{-3}	2.3×10^{-2}	6.7×10^{-2}	0.137	0.216	0.272	0.340	0.182
S_γ	2.1×10^{-3}	4.7×10^{-3}	9.9×10^{-3}	1.6×10^{-2}	2.4×10^{-2}	2.9×10^{-2}	2.9×10^{-2}	0.0192
S_R	6.3×10^{-2}	0.165	0.440	0.912	1.49	2.36	3.84	1.61

Table 4: Raw-sensor sensitivity. Band rms for the affine / non-affine parts: β 0.136 / 0.121, γ 0.0153 / 0.0116, R 1.33 / 0.905. Every sensitivity grows with frequency.

Representation	$(\sigma_1, \sigma_2, \sigma_3)$	κ	$v_{\min}(\beta, \gamma, R)$	joint std (β, γ, R)	fixed-other std
Raw	(3.81, 0.433, 0.0254)	150	(−0.057, −0.998, −0.008)	(3.21, 39.3, 0.406)	(2.31, 22.0, 0.262)
Affine	(3.15, 0.322, 0.0183)	172	(−0.069, −0.998, −0.008)	(4.88, 54.4, 0.533)	(3.09, 27.5, 0.317)
Non-aff.	(2.15, 0.287, 0.0165)	130	(0.041, 0.999, 0.008)	(4.28, 60.6, 0.689)	(3.48, 36.5, 0.466)

Table 5: Frequency-integrated stacked Jacobian (`sensorFRFSensitivity_20260925_130156`). Stds are per unit σ_n . Coupling ratios (joint/fixed) for the raw case: 1.39, 1.79, 1.55.

$R \gg \beta \gg \gamma$, and the weak direction is pure γ in every representation. R is about 84 times and β about 9.5 times more influential than γ (band rms). The raw weak direction (−0.057, −0.998, −0.008) matches the old five-feature map, (−0.034, −0.999, 0.003): the weak γ direction was not caused by the old feature choice, it is already in the full complex sensor response, and no spatial component hides extra γ information. Do not compare κ numerically with the old Table 8 (150 vs 293): different observables and scalings. The geometry is what agrees.

Upper frequency $\omega_{\max}$	$\sigma_{\ln \beta}^{\text{joint}}$	$\sigma_{\ln \gamma}^{\text{joint}}$	$\sigma_{\ln R}^{\text{joint}}$
3.00	908	2.2×10^3	118
3.88	202	667	28.9
4.88	52.1	262	7.82
5.75	17.3	119	2.40
6.62	8.54	76.8	1.18
7.62	4.75	53.7	0.689
8.50	3.21	39.3	0.406

Table 6: Cumulative (frequency-integrated) joint std, raw sensors, all three parameters free. Information arrives mostly at the top of the band and is still growing at 8.5 rad/s.

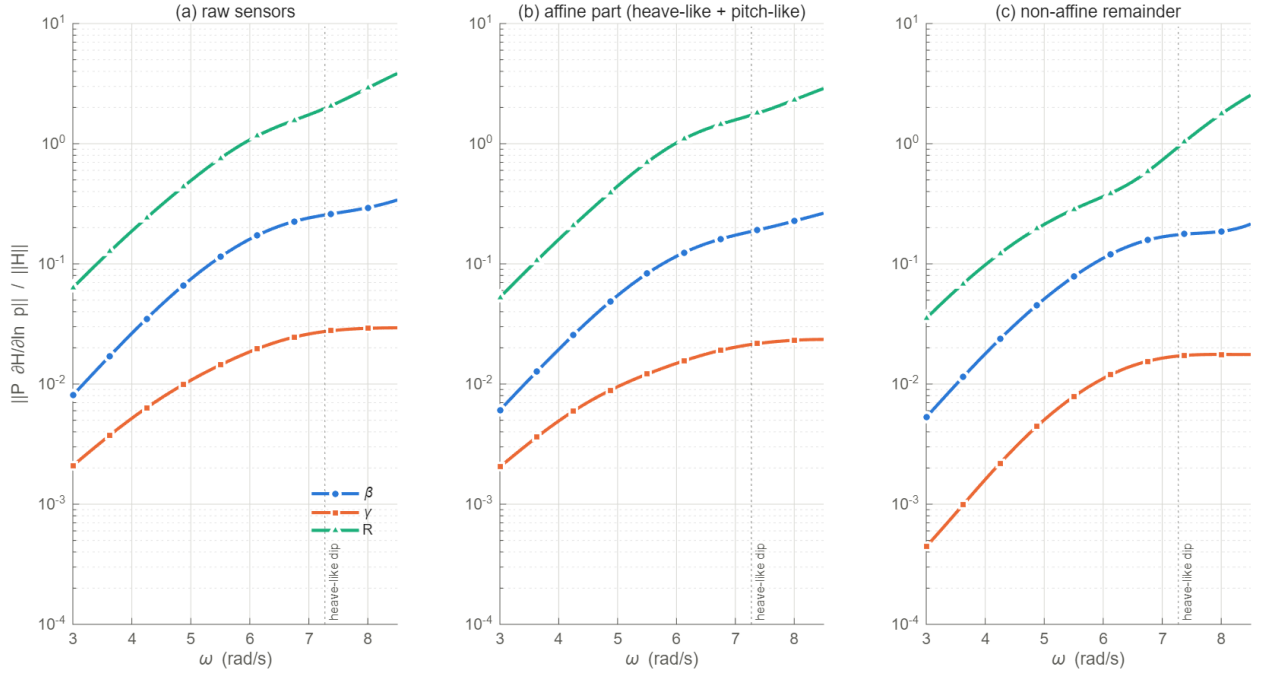


Figure 3: $S_i(\omega)$ for β , γ , R : (a) raw, (b) affine, (c) non-affine.

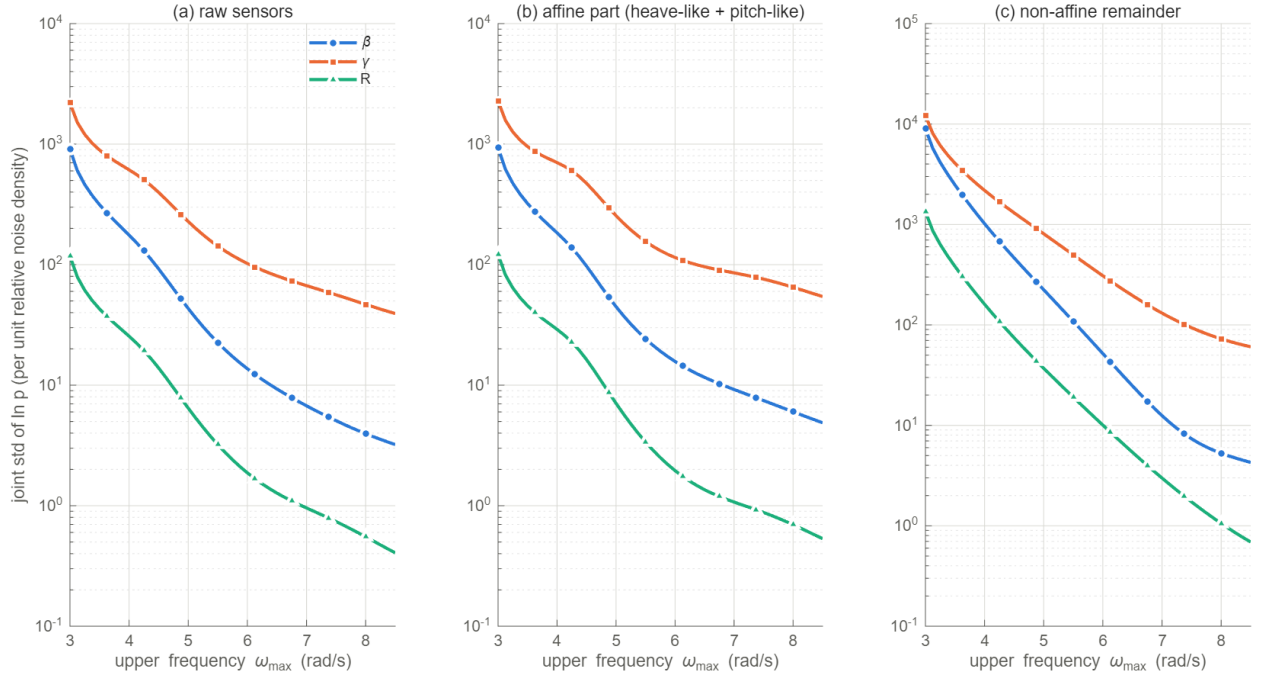


Figure 4: Cumulative joint std versus upper frequency: (a) raw, (b) affine, (c) non-affine.

5.1 Why γ is weak

In the non-dimensional plate condition (Montiel Eq. 2.13d), $(\beta \nabla^4 - \alpha \gamma + 1) \partial_z \phi = \alpha \phi$ on the plate, γ appears only in the mass term $\alpha \gamma$, next to the hydrostatic term 1. At p_0 this term is small throughout the band.

ω (rad/s)	α	$\alpha\gamma_0$	measured S_γ
3.00	1.73	0.0025	0.0021
5.75	6.34	0.0092	0.0164
8.50	13.85	0.0201	0.0294

Table 7: S_γ tracks the mass term $\alpha\gamma$ to within a factor of about two.

γ is weak for physical reasons at this scale. The draught is about 2.7 mm, so $\alpha\gamma \ll 1$. Extending the band to 10 rad/s would raise $\alpha\gamma$ only by $(10/8.5)^2 \approx 1.4$; band extension would sharpen β and R more than γ . Scoped statement: *γ is locally very poorly identifiable from these four complex acceleration FRFs over 3–8.5 rad/s at p_0 ; output-only recovery of γ under this layout is unlikely to be practically useful.* This is not a claim that γ is unrecoverable in general.

5.2 The heave-like dip is not the rigid-disc geometric zero

For a rigid disc in deep water the plane-wave average over the disc is $2J_1(kR)/(kR)$, first zero at $kR = 3.8317$. With $R_{\text{phys}} = 0.383 \times 1.88 = 0.720$ m, $k = 5.32$ m⁻¹ and $\omega = \sqrt{gk \tanh kH} = 7.225$ rad/s. Since $\omega \propto R^{-1/2}$ in deep water, this mechanism predicts $\partial\omega_{\text{dip}}/\partial \ln R \approx -\omega/2 = -3.61$ rad/s.

Rejected. A 21-point local sweep at p_0 and $\pm 5\%$ in each $\ln p$ gave $\omega_{\text{dip}} = 7.273$ rad/s (close to 7.225) but $\partial\omega_{\text{dip}}/\partial \ln R = +2.28$ (opposite sign), $\partial\omega_{\text{dip}}/\partial \ln \beta = +0.253$ and $\partial\omega_{\text{dip}}/\partial \ln \gamma = -0.041$. The frequency agreement is coincidental; a pure wave-following floe would put the c_0 dip near 6.5 rad/s, so that explanation fails too.

6 Decision: fix γ from measured thickness

Ice thickness can be measured before deployment. γ is the draught ratio, $\gamma \propto h$, whereas roughly $\beta \propto Eh^3$: the two share a thickness dependence but are not the same information. Measuring thickness constrains γ and leaves β to identify the effective flexural stiffness. The working inverse vector becomes $\theta = [\ln \beta, \ln R]^T$.

Bias from an error in the fixed γ . Partition the three-parameter information matrix into θ and γ blocks. To first order, an error $\Delta \ln \gamma$ in the assumed value shifts the estimate by

$$\Delta\theta = -F_{\theta\theta}^{-1}F_{\theta\gamma} \Delta \ln \gamma. \quad (10)$$

Quantity	$\ln \beta$	$\ln R$
Joint std, three parameters free	3.21	0.406
Joint std, γ fixed	2.3154	0.2627
σ_n needed for 10% (1σ), γ fixed	$\lesssim 0.043$	$\lesssim 0.38$
Bias per unit $\ln \gamma$ error, $\partial\theta/\partial \ln \gamma$	0.0567	0.0079
Bias from a 20% draught error ($\Delta \ln \gamma = 0.182$)	1.0%	0.14%

Table 8: Effect of fixing γ . For comparison, γ itself would need $\sigma_n \lesssim 0.0025$. Even a 50% draught error biases β by only about 2.3%.

Provisional and reversible. For the rest of the SSI inverse-design study γ is treated as known from pre-deployment thickness/draught; the three-parameter analysis is kept. To be confirmed with the supervisor. The bias estimate is local to p_0 .

7 Transmissibilities (files 6c and 5c)

Transmissibility. With a reference sensor r , the transmissibility of sensor j is the ratio of the two sensor responses,

$$T_j(\omega) = \frac{H_j(\omega)}{H_r(\omega)}, \quad j \neq r. \quad (11)$$

Under the single-common-input model $y_j = H_j u$ (one incident field drives every sensor), the output cross-spectral densities are $S_{jr} = H_j H_r^* S_u$ and $S_{rr} = |H_r|^2 S_u$, so

$$\frac{S_{jr}(\omega)}{S_{rr}(\omega)} = \frac{H_j(\omega)}{H_r(\omega)} = T_j(\omega) : \quad (12)$$

the unknown wave spectrum S_u cancels, and T can be estimated from outputs alone, with no model fit and no resonance required. This holds exactly only for a single common input and a noise-free reference; finite records, sensor noise on the reference and directional or multiple forcing all break it (to be tested in G2). What T discards is the common factor $H_r(\omega)$: the absolute response level and the phase relative to the incident wave.

Consistent noise propagation. To first order, $dT_j = (dH_j - T_j dH_r)/H_r$, i.e. $dT = A d\mathbf{H}$ with $A_{j,j} = 1/H_r$ and $A_{j,r} = -T_j/H_r$. In real form, $\bar{A} = \begin{bmatrix} \text{Re} A & -\text{Im} A \\ \text{Im} A & \text{Re} A \end{bmatrix}$ acts on $[\text{Re } d\mathbf{H}; \text{Im } d\mathbf{H}]$, so $J_T = \bar{A} J_H$ and $\Sigma_T = \sigma_k^2 \bar{A} \bar{A}^T$ with σ_k^2 from Eq. (4). The log variants use $d \ln T = dT/T$: its real part is the log-magnitude and its imaginary part the phase.

`featureInformation.m` implements Eq. (7); its tester (9 groups) confirms that an invertible transform keeps $\lambda = 1$ exactly, 200 random rank-reducing maps keep $\lambda \leq 1$, rescaling changes nothing, an understated noise model is flagged ($\lambda = 4$), and a completely lost direction gives $\lambda = 0$. `runTransmissibilityInformation.m` applied it to the saved file 5b derivatives (no new EMM solves).

Variant	λ_1	λ_2	weak direction (β, R)	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	ratio to FRF
Complex T	0.8339	0.0972	(0.431, 0.902)	3.985	0.784	1.72×, 2.98×
$ T $ only (log-magnitude)	0.3580	0.0107	(0.633, 0.774)	14.82	2.032	6.40×, 7.74×
$\arg T$ only (phase)	0.5131	0.0493	(0.265, 0.964)	4.184	1.159	1.81×, 4.41×
Complex FRF (reference)	1	1	-	2.315	0.263	-

Table 9: Transmissibility information, γ fixed (`transmissibilityInfo_20260925_134954`). Weak directions in reference-std units. The result is invariant to the choice of reference sensor to 4.3×10^{-16} , as it must be: ratios to any other reference are an invertible function of these. Three-parameter case: $\lambda = (0.910, 0.116, 0.089)$, $\sigma_{\ln \gamma} \ 39.3 \rightarrow 115$.

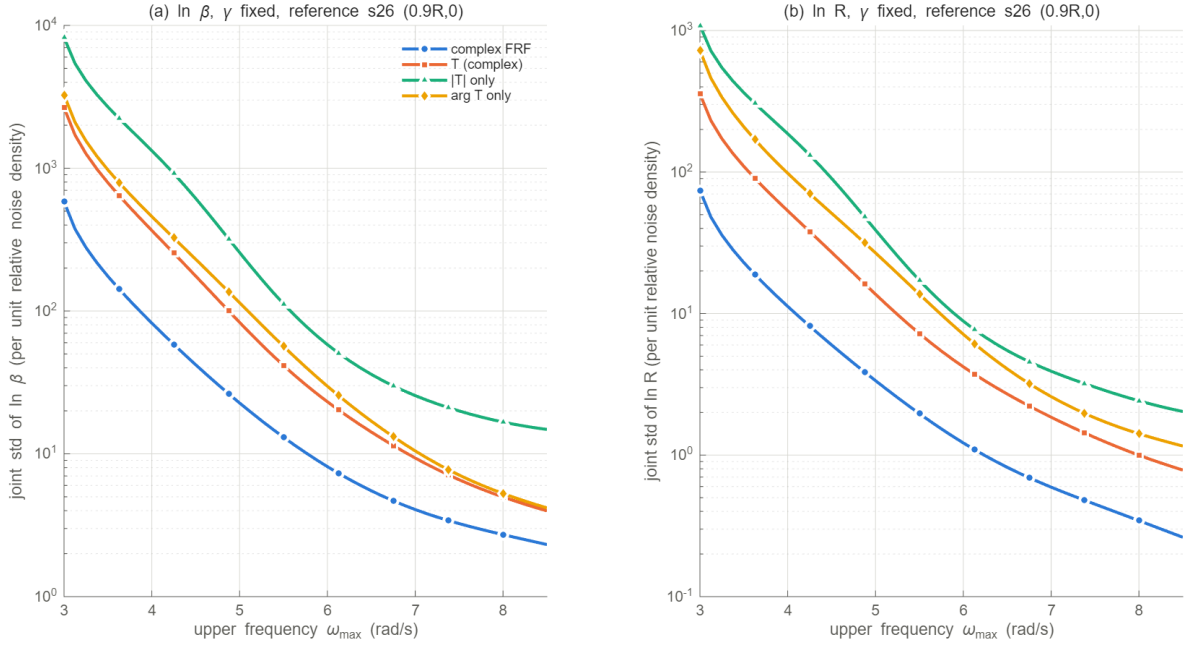


Figure 5: Cumulative joint std of (a) $\ln \beta$ and (b) $\ln R$ versus upper frequency, γ fixed: complex FRF against complex T , $|T|$ only and $\arg T$ only.

Valid, but it loses a direction, mostly R . Transmissibilities keep 83% of the FRF information in one direction and 9.7% in the other; the R std grows $2.98\times$ and β $1.72\times$, and the penalty persists at 8.5 rad/s. Much of the R information therefore lives in the common factor removed by the ratio. For β , almost all transmissibility information is in the relative phase; for R , phase dominates but magnitude adds materially.

8 The output-only information bound (file 5d)

What output-only data contain. With a single common input, the output cross-spectral matrix is

$$S_y(\omega) = \mathbf{H}(\omega)\mathbf{H}(\omega)^H S_u(\omega). \quad (13)$$

At each frequency it is fully described by the transmissibilities T_j and the log auto-spectrum level of the reference,

$$\ell(\omega) = \ln(|H_r(\omega)|^2 S_u(\omega)). \quad (14)$$

The phase of H_r relative to the incident wave is lost for good, so the complex FRF is an unreachable reference for any output-only method. Anything derived from the outputs (SSI poles, damping, mode shapes) is a function of $\{T, \ell\}$ and cannot carry more information. ℓ helps only to the extent that S_u is known, because $|H_r|^2$ and S_u enter as a product.

Joint propagation and nuisance elimination. Per frequency, one transformation

$$[\text{Re } \mathbf{H}; \text{Im } \mathbf{H}] \mapsto [\text{Re } T; \text{Im } T; \ell]$$

is linearised, with

$$d\ell = 2 \text{Re} \left(\frac{dH_r}{H_r} \right) + d \ln S_u. \quad (15)$$

T and ℓ share the noisy reference channel, so their covariance is propagated jointly (treating them as independent would double-count H_r). Unknown spectrum structure is modelled as $\ln S_u = B\eta$ with nuisance parameters η . Their effect is removed by the Schur complement

$$F_{\theta, \text{eff}} = F_{\theta\theta} - F_{\theta\eta} F_{\eta\eta}^{-1} F_{\eta\theta}, \quad (16)$$

computed stably by whitening the Jacobian and projecting the θ columns onto the orthogonal complement of the nuisance columns. Smooth bases B are orthonormal polynomials in normalised frequency. The JONSWAP variant uses the local derivatives of $\ln S_J$ with respect to $(\ln H_s, \ln \omega_p, \ln \gamma_J)$ at a basin-scale nominal $\omega_p = 5.5$ rad/s, $\gamma_J = 3.3$, where

$$S_J(\omega) \propto \omega^{-5} \exp\left[-\frac{5}{4}(\omega_p/\omega)^4\right] \gamma_J^{\exp[-(\omega-\omega_p)^2/(2\sigma^2\omega_p^2)]}, \quad \sigma = 0.07 \ (\omega \leq \omega_p), \ 0.09 \ (\omega > \omega_p). \quad (17)$$

Key	Knowledge of the wave spectrum S_u	λ_1	λ_2	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$
F	Complex FRF (unreachable output-only)	1	1	2.315	0.263
B	S_u exactly known	0.9946	0.4291	2.344 (1.01 $\times$)	0.401 (1.52 $\times$)
C	S_u known up to one scale	0.8836	0.2450	2.835 (1.22 $\times$)	0.513 (1.95 $\times$)
J	JONSWAP form, 3 parameters unknown	0.8537	0.1545	3.377 (1.46 $\times$)	0.629 (2.39 $\times$)
D2	Smooth $\ln S_u$, dimension 2	0.8527	0.1021	3.911	0.765
D4	Smooth $\ln S_u$, dimension 4	0.8367	0.0973	3.982	0.783
D6	Smooth $\ln S_u$, dimension 6	0.8346	0.0972	3.985	0.784
E	$\ln S_u$ arbitrary in every bin	0.8339	0.0972	3.985	0.784
A	Transmissibilities only	0.8339	0.0972	3.985 (1.72 $\times$)	0.784 (2.98 $\times$)

Table 10: Output-only information on $\theta = (\ln \beta, \ln R)$, γ fixed, reference s26 (outputOnlyBound_20260925_140459). Stds per unit σ_n ; ratios to the complex FRF.

All checks passed. E reproduced A to 2.5×10^{-16} (an unknown value per bin absorbs ℓ exactly); A reproduced file 5c; every $\lambda \leq 1$; and information never increased along the nested chain $F \succeq B \succeq C \succeq D2 \succeq D4 \succeq D6 \succeq E$ (smallest eigenvalue of each difference $\geq -1.5 \times 10^{-18}$ relative).

With an unknown sea spectrum, output-only information collapses to transmissibilities. Allowing $\ln S_u$ only a constant-plus-linear freedom (D2) already removes almost all of the extra information in ℓ : the θ -dependence of $|H_r|^2$ is smooth enough to be mimicked by a spectral tilt. Knowing the spectrum is what pays: exact knowledge (B) brings β back to FRF level and halves the R penalty; knowledge up to a scale (C) more than doubles λ_2 over transmissibilities.

Scope. This is an output-only *second-order* bound under the single-common-input model and the same local noise surrogate as files 5b/5c. The J row is optimistic: it assumes the sea really is JONSWAP around the nominal peak, and a wrong shape would turn information into bias. Real finite-record covariance comes in G2.

8.1 Consequences for SSI

Poles are demoted to an optional comparison. Interpreting output-only SSI poles as physical system poles relies on the unmeasured excitation being broadband enough that its colour does not masquerade as system dynamics. Table 10 shows why this matters here: a smooth unknown S_u absorbs the auto-spectrum information, so any pole method that appears to recover more than the transmissibility information from coloured data deserves suspicion, since it may be fitting excitation colour rather than floe dynamics. Global pole fitting is therefore removed from the critical path and kept as a later experiment: under coloured excitation, do SSI poles track the floe or the sea?

9 Working architecture after Friday

Scenario O (output-only, unknown wave spectrum). The core observable is the full frequency-resolved complex transmissibility vector

$$f_T(\theta) = [\text{Re } T(\omega_1); \text{Im } T(\omega_1); \dots; \text{Re } T(\omega_N); \text{Im } T(\omega_N)], \quad \theta = (\ln \beta, \ln R), \quad (18)$$

with the pipeline

$$\text{EMM} \longrightarrow T(\omega; \beta, R) \longrightarrow \hat{T} \text{ (from time series)} \longrightarrow \Sigma_T \longrightarrow (\beta, R) \text{ inversion}. \quad (19)$$

No band averaging: averaging is another irreversible compression. If the dimension becomes inconvenient, compress afterwards against the information metric, for example with the score-like statistic $J_T^T \Sigma_T^{-1} (f_T - f_{T,0})$.

Scenario W (measured or constrained wave spectrum). With a wave probe giving S_u , the usable feature becomes $\{T, \ell\}$ (row B). If the probe is also synchronised with the floe sensors and the propagation from probe to floe is known, an input-output FRF may be estimable, moving beyond the output-only problem. For field deployment an open-water reference needs care: MIZ forcing may be directional and spatially varying.

10 Files built on Friday

File (folder)	Role	Outcome
stochasticSensorSynthesis.m (SSI/ValidationExperiments/) + tester	Shared-input frequency-domain synthesis	10 groups pass
level4A_broadbandKnownOscillator.m (InverseMapping/) + tester	Known-oscillator SSI gate	Section 3
computeSensorFRF.m (InverseMapping/) + tester	EMM sensor FRFs, displacement or acceleration	10 groups pass
runSensorFRFReconnaissance.m (.../SSITwin/)	FRF structure at p_0 (file 5)	Section 4
runSensorFRFSensitivity.m (.../SSITwin/)	FRF-level sensitivity (file 5b)	Section 5
featureInformation.m (InverseMapping/) + tester	Generalised-eigenvalue information comparison (file 6c)	9 groups pass; frozen
runTransmissibilityInformation.m (.../SSITwin/)	Transmissibility information (file 5c)	Section 7
runOutputOnlyInformationBound.m (.../SSITwin/)	Output-only bound (file 5d)	Section 8

Table 11: Friday’s files. .../SSI Twin/ is Inverse Mapping/Parameter Inversion/SSI Twin/; results are in Inverse Mapping/Parameter Inversion/Results/.

11 Weekend plan

11.1 Saturday 26 September 2026: the estimator stage (gate G2)

Goal: can the ideal transmissibilities be estimated reliably from finite stochastic time series, and with what bias and covariance?

File	Folder	Purpose
ssiFeatureVector.m + tester	InverseMapping/	Deterministic forward map $f_T(\beta, R \mid \gamma \text{ fixed})$: full complex transmissibility vector on a fixed grid, via <code>computeSensorFRF</code>
extractTransmissibility.m + tester	SSI/	Estimate $\hat{T}_j = \hat{S}_{jr}/\hat{S}_{rr}$ from time series (Welch cross-spectra), with coherence diagnostics and a covariance estimate
runTransmissibilityEstimatorTest.m	.../SSITwin/	G2: bias and Σ_T versus record length for (i) white common input, (ii) JONSWAP common input, (iii) sensor noise, (iv) combinations

Table 12: Saturday files.

Questions to answer:

1. Does coloured common input leave $\hat{T}$ unbiased while changing its variance (low-energy bins)?

2. How strongly does sensor noise bias the reference auto-spectrum S_{rr} , and hence $\hat{T}$?
3. How do bias and covariance scale with record length?
4. Does the empirical Σ_T reduce the (β, R) information relative to the surrogate of Eq. (4)?

Gate G2: $\hat{T}$ is recovered with bias small relative to its scatter under white and JONSWAP input with realistic noise, and Σ_T is measured. The single-common-input assumption stays explicit; directional or multiple forcing is outside this scope.

11.2 Sunday 27 September 2026: covariance-weighted information and twin inversion (gate G3)

Goal: do output-only transmissibility estimates recover (β, R) ?

File	Folder	Purpose
runTransmissibilityCovariance.m	.../SSITwin/	Recompute $F = J_T^T \Sigma_T^{-1} J_T$ with the estimated Σ_T ; compare with the file 5c/5d surrogate via <code>featureInformation</code>
runTransmissibilityTwinInversion.m	.../SSITwin/	$\hat{T}$ from twin time series $\rightarrow$ <code>solveInverse</code> over $(\ln \beta, \ln R) \rightarrow \hat{\theta}$ against truth, several realisations
Scenario W (if time)	-	Repeat with $\{T, \ell\}$ and known S_u : the wave-probe case

Table 13: Sunday files. `solveInverse`, `inverseResidual` and `probeValidRegion` are reused unchanged; only the parameter count drops from three to two.

Gate G3: output-only transmissibility estimates with realistic finite-record covariance recover (β, R) consistently, with errors in line with the covariance-weighted prediction.

11.3 Deferred

Global pole fitting and the coloured-input SSI experiment (optional comparison); common-random-number clean-up of Level 4A Stage 5; EMM truncation revalidation above 8.5 rad/s (would sharpen β and R , not γ); sensor re-optimisation, now against $\theta = (\ln \beta, \ln R)$.

12 Supervisor discussion points

1. **Fix γ from thickness/draught.** γ is the weak physical direction (Tables 5, 7), and a 20% draught error biases β by about 1% and R by about 0.14% (Table 8).
2. **Measure the incident wave in the tank campaign.** A wave probe giving S_u moves the output-only ceiling from row A to row B of Table 10; a synchronised probe may allow a direct input-output FRF.

These are experimental-design choices, and they change the achievable accuracy more than any feature design.